Review Article

Chemical Components and Biological Activities in *Lilii Bulbus* Based on Scientometrics : A Review

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ABSTRACT

The inheritance and utilization of "homology of medicine and food" have become the focus of the society and the market. In order to avoid the singularity of traditional reviews, the scientometric method (software "CiteSpace") was used to conduct visual knowledge graph analysis and summarize the existing Chinese and English literature on *Lilii Bulbus* at home and abroad, and predict the research hotspots of *Lilii Bulbus*. There are 235 chemical components in *Lilii Bulbus*, including 111 steroidal saponins, 17 polysaccharides, 14 flavonoids, 21 steroidal alkaloids, 53 phenylpropenoid glycerides, and 19 other chemical components. The biological activities of *Lilii Bulbus* from extraction methods, compounds, models, and effective concentrations were summarized. In addition, scientometric software analysis shows that the research on *Lilii Bulbus* can be classified into five categories: chemical components, biological activities, species origin, quality evaluation, and methods of extraction and purification. It is worth noting that the identification and assessment of chemical components (like steroidal glycosides) of *Lilii Bulbus* and their biological activities (such as anti-inflammatory and antioxidant effects) are the current research foci for *Lilii Bulbus* research. To sum up, *Lilii Bulbus*, which is homologous in medicine and food, has great potential in the treatment of cancer and inflammation and can also be used as a cosmetic raw material of natural origin to deeply explore the efficacy of cosmetics and increase the added value of products.

Introduction

Lilii Bulbus is one of the important representative species of dual-purpose Chinese traditional medicine and health products approved by the Ministry of Health [1]. In general, *Lilii Bulbus* (the subterranean fleshy scale leaves) are the edible parts of *Lilium lancifolium* Thunb., *Lilium brownii F. E. Brown var. viridulum* Baker, and *Lilium davidii var. unicolor Cotton* [2]. The main chemical components of *Lilii Bulbus* are regarded as steroidal saponins[3], polysaccharides [4], flavonoids [5], steroidal alkaloids [6], and phenols [7], and biological activities are considered to be anti-inflammatory [8], antioxidant [9], anti-tumor [10], antidepressant [11], hypoglycemic [12], and antibacterial [13]. Although a large amount of academic literature related to the chemical components and biological activities of *Lilii Bulbus* has been published in English or Chinese in the last 25 years, there is no up-to-date

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systematic summary and prediction of its research hotspots.

Yu adopts the scientometric analysis method to assess the current state and explore the development trends of carbon emission trading domain based on the literature data retrieved from Web of Science [14]. Liu utilized bibliometric methods to describe the development of AI in food safety, including performance analysis, science mapping, and network analysis using CiteSpace [15]. Sun Yaqing [16] employed CiteSpace to describe the trends and comparisons of literature outputs, "star" papers and hotspots, illustrating the emerging trend of research on PAHs in food. So far, it has been widely used to detect research trends and frontiers, the development of the entire discipline and the promotion of interaction between different fields.

Therefore, CiteSpace V6.2.R4 software [17] was used to conduct visualized knowledge mapping with countries, institutions, authors, journals, references, and keywords of *Lilii Bulbus* literature retrieved from Web of Science (WOS) databases and China National Knowledge Infrastructure (CNKI). The framework of this review and the overall research process are as shown in Figure 1. In this study, the chemical components and biological activities of *Lilii Bulbus* were inducted and summarized and the scientometric analysis was used to present the hot topics and dynamic frontiers from a macro perspective [18], providing reference for subsequent research directions, with a view to promoting further in-depth research on *Lilii Bulbus*.

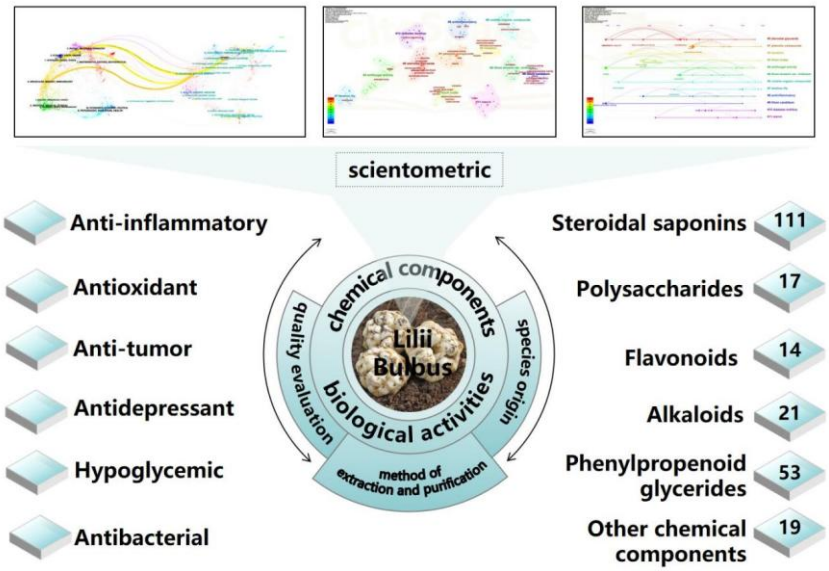


Figure 1 The framework of the review and the overall research process

Data Sources

In this study, the English literature was retrieved on July 15, 2023, from the WOS database created by Clarivate Analytics. The statistical caliber of the data [19] is described as follows: the database selected was searched in the Web of Science Core Collection, and the editions selected were Science Citation Index Expanded (SCI-E) and Conference Proceedings Citation Index-Science (CPCI-S) The topic-based search strategy was developed based on the keywords involved, and the search formula was: Topic = (*Lilium* OR *Lily* OR *Lilii Bulbus*). The time span was from January 1, 1998 to July 15, 2023, and the document types included Article, Meeting Abstract, Review Article, with the language being English. English records were retrieved from WOS-core on July 15, 2023, with the string: TS=(*Lilium* OR "*Lilii Bulbus*" OR *Lily*) AND DT=(Article OR Review OR Meeting Abstract) (1998-2023), yielding 1,247 hits. After duplicate

removal and two-round screening (title/abstract & full text, κ=0.89), 143 records remained; exclusion reasons are listed in Supplementary Table S1.

The Chinese literature was retrieved on July 15, 2023, from China National Knowledge Infrastructure (CNKI) database. The advanced search interface was entered and the search criteria was inputed [20]: Topic "Baihe (*Lilii Bulbus* in Chinese expressions)", NOT Topic "Ye-Baihe-Jian (monocrotaline in Chinese expressions)", selected the literature category as "science and technology", selected the discipline area as "traditional Chinese medicine". The time span was from January 1, 1998 to July 15, 2023, and the type of literature was unlimited. By excluding duplicate publications and screening out literature that did not conform to the topics related to *Lilii Bulbus* research in the fields of chemical components and biological activities, 318 valid literature were finally included, and further analysis was performed with scientometric software tools. The subject "Ye-Baihe-Jian" is a chemical component unrelated to *Lilii Bulbus*. "Ye-Baihe-Jian" and "Baihe" overlap in words of the Chinese expression, so literature with the subject "Ye-Baihe-Jian" was excluded from the search criteria.

Statistical Analysis

The characteristics of *Lilii Bulbus* research literature output according to the above literature dataset were used to quantitatively analyze and identify through different metrics (chronological distribution), and then objectively review the current development of *Lilii Bulbus* research in the fields of chemical components and biological activities. Based on scientometric analysis, the research subjects' cooperation (country, institution and author), co-citation (journals), keyword co-occurrence (co-occurrence, clustering, and highlighting), and research hotspot trends were analyzed, and visual scientific knowledge maps (clustering map, timeline view, and biplot overlay) were displayed [21]. The results obtained from the above statistical analysis were interpreted, and then combined with the full text of literature, in order to identify and reflect the dynamic processes of the basic framework, knowledge mobility, frontier shifts, and evolution in the fields of chemical components and biological activity research of *Lilii Bulbus*.

CiteSpace V6.2.R4 running parameters were set as : Time Slicing from 1998 to 2023, #Year Per Slice set to "1", Node Types including Author, Country, Institution, Keyword, Reference, Cited Author, and Cited Journal, Selection Criteria was g-index. The mapping pruning algorithm was applied with Pathfinder, pruning sliced networks, and pruning the merged network under pruning, in order to simplify the visual mapping of the output [22]. The size of the font indicates the frequency of the term, the line between the nodes represents the cooperation [23-25], and the color of the line between the nodes determines the year of the publication; a thicker line indicates the stronger cooperation between the two terms [26-28]. In this way, a visual analysis of the trend of the dual-map overlay of journals, co-cited reference knowledge graph, author collaborative knowledge graph, and co-occurrence, clustering, and emergent knowledge graph of keywords was performed.

Results and Discussion

The Chemical Components of Lilii Bulbus

The article was summarized 111 steroidal saponins, 17 polysaccharides, 14 flavonoids, 21 alkaloids, 53 phenylpropenoid glycerides, and 19 other chemical components of *Lilii Bulbus*, and presented in Tables 1-6.

The Biological Activities of Lilii Bulbus

To date, researchers have carried out a series of studies on the chemical components of *Lilii Bulbus*, such as saponins and polysaccharides, and found that they have the effects of anti-inflammatory, antioxidant, anti-tumor, antidepressant, hypoglycemic, antibacterial, among others. These biological activities from extraction methods, compounds, models, effective concentrations were summarized, as presented in Table 7.

Table 1 Steroidal saponins of *Lilii Bulbus*

No.	Compound name	Molecular formula	Source	[Ref.]
1	(25*S*)-26-*O*-*β*-*D*-glucopyranosyl-22,25-epoxyfurost-5*α*-3*β*,26-diol 3-*O*-[*α*-*L*-rhamnopyranosyl-(1→2)]-*β*-*D*-glucopyranoside	~	*L.brownii*	[12]
2	(25*R*)-27-*O*-[(*S*)-3-hydroxy-3-methylglutaryl]-spirost-5*α*-3*β*,27-diol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside	~	*L.brownii*	[12]
3	Isonarthogenin 3-*O*-{*O*-*β*-*D*-glucopyranosyl-(1→3)-*O*-*α*-*L*-rhamnopyranosyl-(1→2) -*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside}	$C_{51}H_{82}O_{23}$	*L.candidum*	[29]
4	(25*R*,26*R*)-26-methoxyspirost-5-ene-3*β*,17*α*-diol 3-*O*-{*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-*β*-*D*-glucopyranoside}	$C_{40}H_{64}O_{14}$	*L.candidum*	[29]
5	(25*R*,26*R*)-26-methoxyspirost-5-ene-3*β*,17*α*-diol 3-*O*-{*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside}	$C_{46}H_{74}O_{19}$	*L.candidum*	[29]
6	(25*R*,26*R*)-26-methoxyspirost-5-ene-3*β*,17*α*-diol 3-*O*-{*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[6-*O*-acetyl-*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside}	$C_{48}H_{76}O_{20}$	*L.candidum*	[29]
7	(25*R*)-spirost-5-ene-3*β*,27-diol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-*β*-*D*-glucopyranoside	~	*L.candidum*	[29]
8	(25*R*)-26-*O*-*β*-*D*-glucopyranosyl-22-*O*-methylfurost-5-ene-3*β*,22ξ,26-triol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-*β*-*D*-glucopyranoside	$C_{46}H_{76}O_{18}$	*L.candidum*	[29]
9	(25*S*)-17*α*,27-dihydroxyspirost-5-en-3*β*-yl *O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{45}H_{72}O_{19}$	*L.speciosum*	[3]
10	(25*S*)-17*α*,27-dihydroxyspirost-5-en-3*β*-yl *O*-*β*-*D*-glucopyranosyl-(1→3)-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside.	$C_{51}H_{82}O_{24}$	*L.speciosum*	[3]
11	(25*S*,26*R*)-26-methoxyspirost-5-en-3*β*-yl *O*-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside	$C_{40}H_{64}O_{13}$	*L.speciosum*	[3]
12	(25*R*)-17*α*-hydroxy-27-[(3-hydroxy-3-methylglutaroyl)oxy]-spirost-5-en-3*β*-yl *O*-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside.	$C_{45}H_{70}O_{18}$	*L.speciosum*	[3]
13	(25*R*)-27-[(3-hydroxy-3-methylglutaroyl)oxy]-spirost-5-en-3*β*-yl O-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→6)]-*β*-*D*-glucopyranoside	$C_{51}H_{80}O_{22}$	*L.speciosum*	[3]
14	(25*R*)-[[3-(*β*-*D*-glucopyranosyl)oxy-3-methylglutaroyl]oxy]-spirost-5-en-3*β*-yl O-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{57}H_{90}O_{27}$	*L.speciosum*	[3]
15	(25*R*)-27-hydroxyspirost-5-en-3*β*-yl O-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside	$C_{45}H_{72}O_{19}$	*L.speciosum*	[3]
16	(25*R*)-17*α*-hydroxy-3-[(*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranosyl)oxy]-5*α*-spirostan-6-one	$C_{45}H_{76}O_{19}$	*L.speciosum*	[3]
17	Diosgenin 3-*O*-*β*-*D*-glucopyranosyl-(1→2)-*β*-*D*-glucopyranosyl-(1→4) -*β*-*D*-galactoside (funkioside D)	$C_{45}H_{72}O_{18}$	*L.callosum*	[30]

No.	Compound name	Molecular formula	Source	[Ref.]
18	Diosgenin 3-*O*-[*β*-*D*-glucopyranosyl-(1→2)]-[*β*-*D*-xyl-(1→3)]-*β*-*D*-glucopyranosyl-(1→4)-*β*-*D*-galactoside (aspidistrin)	$C_{50}H_{80}O_{22}$	*L.callosum*	[30]
19	(25*R*)-3*β*,17*α*,27-triol-spirostan-6-one 3-*O*-*α*-*L*-rha-(1→2)-[*β*-*D*-glucopyranosyl-(1→6)]-*β*-*D*-glucopyranoside (callosum A)	$C_{45}H_{72}O_{20}$	*L.callosum*	[30]
20	(25*R*)-2*α*,3*β*-*D*ihydroxy-spirost-5-en-1 *β*-*O*-*α*-*L*-rhamnopyranosyl-(1→2) -*β*-*D*-xyloside (atropuroside A)	$C_{38}H_{60}O_{13}$	*L.callosum*	[30]
21	(25*R*)-2*α*,3*β*,17*α*-trihydroxy-spirost-5-en-1*β*-*O*-*β*-*D*-xyloside (atropuroside C)	$C_{32}H_{50}O_{10}$	*L.callosum*	[30]
22	Dioscin	$C_{45}H_{72}O_{16}$	*L.pumilum DC*	[31]
23	(25*R*)-5*α*-spirostan-3*β*,17*α*-diol 3-*O*-*β*-*D*-xylopyranosyl-(1→4) -[*α*-*L*-arabinopyranosyl-(1→6)]-*β*-*D*-glucopyranoside	$C_{43}H_{70}O_{17}$	*L.pumilum DC*	[31]
24	(25*R*)-3*β*-hydroxy-5*α*-spirostan-6-one 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside	$C_{39}H_{62}O_{13}$	*L.pumilum DC*	[31]
25	(25*R*)-3*β*-hydroxy-5*α*-spirostan-6-one 3-*O*-*α*-*L*-arabinopyranosyl-(1→6)-*β*-*D*-glucopyranoside	$C_{38}H_{60}O_{14}$	*L.pumilum DC*	[31]
26	(25*R*)-3*β*,17*α*,27-triol-spirostan-6-one 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside (pumilum A)	$C_{39}H_{62}O_{15}$	*L.pumilum DC*	[31]
27	Daucosterol	$C_{35}H_{60}O_6$	*L.lancifolium*	[32]
28	stigmassterol	$C_{29}H_{40}O$	*L.lancifolium*	[32]
29	*β*-stigmassterol	$C_{29}H_{50}O$	*L.lancifolium*	[32]
30	27-*O*-[(3S)-3-*O*-*β*-*D*-glucopyranosyl 3-methylglutaroyl] isonarthogenin 3-*O*-[*α*-*L*-rhamnopyranosyl-(1→2)]-*β*-*D*-glucopyranoside	$C_{51}H_{80}O_{22}$	*L.brownii*	[33]
31	26-*O*-*β*-*D*-glucopyranosylnuatigenin 3-*O*-*β*-*D*-glucopyranoside	$C_{39}H_{62}O_{14}$	*L.brownii*	[33]
32	26-*O*-*β*-*D*-glucopyranosylnuatigenin 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside	$C_{45}H_{74}O_{18}$	*L.brownii*	[33]
33	26-*O*-*β*-*D*-glucopyranosylnuatigenin 3-*O*-{*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→6)]}-*β*-*D*-glucopyranoside	$C_{51}H_{82}O_{23}$	*L.brownii*	[33]
34	*O*-*β*-*D*-glucopyranosylnuatigenin 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{51}H_{82}O_{23}$	*L.brownii*	[33]
35	26-*O*-*β*-*D*-glucopyranosylnuatigenin	$C_{33}H_{52}O_9$	*L.brownii*	[33]
36	26-*O*-*β*-*D*-glucopyranosyl-3*β*, 26-dihydroxy-5-choleslen-16, 22-dioxo-3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside	$C_{45}H_{72}O_{18}$	*L.brownii*	[33]
37	*O*-*β*-*D*-glucopyranosyl-3*β*, 26-dihydroxy-cholestan-16, 22-dioxo-3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside	$C_{45}H_{74}O_{18}$	*L.brownii*	[33]
38	26-*O*-[*β*-*D*-glucopyranosyl-(1→2)]-*β*-*D*-glucopyranosylnuatigenin 3-*O*-[*α*-*L*-rhamnopyranosyl-(1→2)]-*β*-*D*-glucopyranoside	$C_{51}H_{82}O_{23}$	*L.brownii*	[33]

No.	Compound name	Molecular formula	Source	[Ref.]
39	(25*R*)-3*β*,17*α*-dihydroxy-5*α*-spirostan-6-one 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside	$C_{39}H_{62}O_{15}$	*L.brownii*	[33]
40	(24*S*,25*S*)-3*β*,17*α*,24-trihydroxy-5*α*-spirostan-6-one 3-*O*-[*α*-*L*-rhamnopyranosyl-(1→2)]-*β*-*D*-glucopyranoside	$C_{39}H_{62}O_{15}$	*L.brownii*	[33]
41	(20*R*,22*R*)-3*β*, 20, 22-trihydroxy-5*α*-cholestan-6-one 3-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{39}H_{66}O_{15}$	*L.brownii*	[33]
42	27-*O*-(3-hydroxy-3-methylglutaroyl) -spirost-5-ene-3*β*, 27-diol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	~	*L.brownii*	[34]
43	27-*O*-(3-hydroxy-3-methylglutaroyl) isonarthogenin 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside	~	*L.brownii*	[34]
44	27-*O*-(3-hydroxy-3-methylglutaroyl) isonarthogenin 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2) -*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{51}H_{80}O_{22}$	*L.brownii*	[34]
45	(25*S*)-spirost-5-ene-3*β*,27-diol-3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside (deacylbrownioside /lilioglycoside E)	$C_{39}H_{62}O_{13}$	*L.brownii*	[34]
46	26-*O*-*β*-*D*-glucopyranosylnuatigenin 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{51}H_{84}O_{23}$	*L.brownii*	[34]
47	26-*O*-*β*-*D*-glucopyranosyl-(25R) -furost-5-en-3*β*, 22ξ-diol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{51}H_{84}O_{23}$	*L.hansonii*	[35]
48	26-*O*-*β*-*D*-glucopyranosyl-(25R) -5*α*-furostan-3*β*, 22ξ-diol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{51}H_{86}O_{23}$	*L.hansonii*	[35]
49	(25*R*)-spirost-5-en-3*β*,12*α*-diol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{45}H_{72}O_{18}$	*L.hansonii*	[35]
50	(25*R*)-5*α*-spirostan-3*β*,12*α*-diol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{45}H_{74}O_{18}$	*L.hansonii*	[35]
51	(25*R*)-3*β*,17*α*-diol-5*α*-spirostan 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	~	*L.hansonii*	[35]
52	22-*O*-methyl-26-*O*-*β*-*D*-glucopyranosyl-(25R)-furost-5-ene-3*β*,22ξ,26-triol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2) -*O*-[*β*-*D*-xylopyranosyl-(1→3)]-*β*-*D*-glucopyranoside	$C_{51}H_{84}O_{22}$	*L.longiflorum*	[36]
53	22-*O*-methyl-26-*O*-*β*-*D*-glucopyranosyl-(25R)-furost-5-ene-3*β*,22ξ,26-triol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2) -*O*-[*α*-*L*-arabinopyranosyl-(1→3)]-*β*-*D*-glucopyranoside	$C_{51}H_{84}O_{22}$	*L.longiflorum*	[36]
54	(25*S*)-spirost-5-ene-3*β*,27-diol 3-*O*-{*O*-*α*-*L*-rhamnopyranosyl-(1→2) -*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside}	$C_{45}H_{72}O_{18}$	*L.longiflorum*	[36]
55	(25S)-spirost-5-ene-3*β*,27-diol 3-*O*-{*O*-*α*-*L* rhamnopyranosyl-(1→2)-*O*-[*α*-*L*-arabinopyranosyl-(1→3)]-*β*-*D*-glucopyranoside}	$C_{44}H_{70}O_{17}$	*L.longiflorum*	[36]

No.	Compound name	Molecular formula	Source	[Ref.]
56	(25*R*)-27-*O*-[(*S*)-3-hydroxy-3-methylglutaryl]-spirost-5-ene-3*β*,27-diol 3-*O*-{*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*α*-*L*-arabinopyranosyl-(1→3)]-*β*-*D*-glucopyranoside}	$C_{50}H_{78}O_{21}$	*L.longiflorum*	[36]
57	(25*R*)-27-*O*-[(*S*)-3-hydroxy-3-methylglutaryl]-spirost-5-ene-3*β*,27-diol 3-*O*-{*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(l→4)]-*β*-*D*-glucopyranoside}	~	*L.longiflorum*	[36]
58	22-*O*-methyl-26-*O*-*β*-*D*-glucopyranosyl-(25R)-furost-5-ene-3*β*,22ξ,26-triol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{52}H_{86}O_{23}$	*L.regale*	[37]
59	(25*S*)-spirost-5-ene-3*β*,17*α*,27-triol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→2)-*O*-*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{51}H_{82}O_{24}$	*L.regale*	[37]
60	(25*R*)-spirost-5-en-3*β*-ol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→3)]-*β*-*D*-glucopyranoside (lilioglycoside G /gracillin)	$C_{45}H_{72}O_{17}$	*L.regale*	[37]
61	(25*R*)-27-*O*-[(S)-3-hydroxy-3-methylglutaroyl]-spirost-5-ene-3*β*,27-diol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→3)]-*β*-*D*-glucopyranoside	$C_{51}H_{80}O_{22}$	*L.regale*	[37]
62	(25*R*)-22-*O*-methyl-26-*O*-*β*-*D*-glucopyranosyl-furost-5-ene-3*β*,22ξ,26-triol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glu-(1→3) -*β*-*D*-glucopyranoside (trigofoenosideD-l)	$C_{52}H_{86}O_{23}$	*L.regale*	[37]
63	22-*O*-methyl-26-*O*-acetyl-(25*R*) -5*α*-furost-3*β*,17*α*,22*α*,26-tetraol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2) -*β*-*D*-glucopyranoside (pardarinoside B)	$C_{42}H_{70}O_{15}$	*L.pardarinum*	[38]
64	22-*O*-methyl-26-*O*-acetyl-(25*R*)-5*α*-furost-3*β*,17*α*,22*α*,26-tetraol 3-*O*-[*α*-*L*-rhamnopyranosyl-(1→2)][*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside (pardarinoside D)	$C_{48}H_{82}O_{20}$	*L.pardarinum*	[38]
65	22-*O*-methyl-26-*O*-acetyl-(25*R*)-5*α*-furost-3*β*,17*α*,22*α*,26-tetraol 3-*O*-[*α*-*L*-rhamnopyranosyl-(1→2)][*α*-*L*-arabinopyranosyl-(1→3)]-*β*-*D*-glucopyranoside (pardarinoside G)	$C_{47}H_{80}O_{19}$	*L.pardarinum*	[38]
66	*O*-methyl-26-*O*-acetyl-(25*R*)-5*α*-furost-3*β*,14*α*,17*α*,22*α*,26-pentaol 3-*O*-[*α*-*L*-rhamnopyranosyl-(1→2)][*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside (pardarinoside C)	$C_{48}H_{82}O_{21}$	*L.pardarinum*	[38]
67	22-*O*-methyl-26-*O*-acetyl-(25*R*) -5*α*-furost-3*β*,14*α*,17*α*,22*α*,26-pentaol 3-*O*-[*α*-*L*-rhamnopyranosyl-(1→2)] -*β*-*D*-glucopyranoside (pardarinoside A)	$C_{42}H_{70}O_{16}$	*L.pardarinum*	[38]
68	*O*-methyl-26-*O*-acetyl-(25*R*)-5*α*-furost-3*β*,14*α*,17*α*,22*α*,26-pentaol 3-*O*-[*α*-*L*-rhamnopyranosyl-(1→2)][*α*-*L*-arabinopyranosyl-(1→3)]-*β*-*D*-glucopyranoside (pardarinoside F)	$C_{47}H_{80}O_{20}$	*L.pardarinum*	[38]
69	(25*R*)-spirost-3*β*,17*α*,21-triol 3-*O*-[*α*-*L*-rhamnopyranosyl-(1→2)][*α*-*L*-arabinopyranosyl-(1→3)]-*β*-*D*-glucopyranoside (pardarinoside E)	$C_{45}H_{74}O_{18}$	*L.pardarinum*	[38]

No.	Compound name	Molecular formula	Source	[Ref.]
70	(25*S*)-spirost-5-ene-3*β*,27-diol 3-*O*-*β*-*D*-glucopyranoside (lilioglycoside B)	$C_{33}H_{52}O_9$	*L.regale*	[39]
71	(25*S*)-spirost-5-ene-3*β*,27-diol 3-*O*-{[O-*α*-*L*-rhamnopyranosyl-(1→2)]-[*O*-*β*-*D*-glucopyranosyl-(l→3)-*β*-*D*-glucopyranoside} (lilioglycoside H)	$C_{45}H_{72}O_{19}$	*L.regale*	[39]
72	(25*R*)-spirost-5-ene-3*β*, 27-diol 3-*O*-{[O-*α*-*L*-rhamnopyranosyl-(1→2)-[*β*-*D*-glucopyranosyl-(1→3)]-*β*-*D*-glucopyranoside} 27-[(S) -3-hydroxy-3-methylglutarate)](lilioglycoside I)	$C_{51}H_{80}O_{23}$	*L.regale*	[39]
73	(25*R*)-27-(3-hydroxy-3-methylglutarate)-spirost-5-ene-3*β*,27-diol 3-*O*-*β*-*D*-glucopyranoside (lilioglycoside C)	$C_{39}H_{60}O_{13}$	*L.regale*	[39]
74	(25*S*)-spirost-5-ene-3*β*,17*α*,27-triol 3-*O*-{*O*-*β*-*D*-glucopyranosyl-(1→2)-*O*-*β*-*D*-glucopyranosyl-(1→4)-*β*-*D*-glucopyranoside}	$C_{45}H_{72}O_{20}$	*L.martagon*	[40]
75	(25*S*)-5*α*-spirostane-3*β*,17*α*,27-triol 3-*O*-{*O*-*β*-*D*-glucopyranosyl-(1→2)-*O*-*β*-*D*-glucopyranosyl-(1→4)-*β*-*D*-glucopyranoside}	$C_{45}H_{74}O_{20}$	*L.martagon*	[40]
76	(25*S*)-27-hydroxyspirost-5-en-3*β*-yl *O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→6)]-*β*-*D*-glucopyranoside	$C_{45}H_{72}O_{18}$	*L.candidum*	[41]
77	(25*R*,26*R*)-26-methoxyspirost-5-en-3*β*-yl *O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→6)]-*β*-*D*-glucopyranoside	$C_{46}H_{74}O_{18}$	*L.candidum*	[41]
78	(25*R*,26*R*)-17*α*-hydroxy-26-methoxyspirost-5-en-3*β*-yl *O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→6)]-*β*-*D*-glucopyranoside	$C_{46}H_{74}O_{19}$	*L.candidum*	[41]
79	(25*R*)-spirost-5-en-3*β*-yl O-*α*-*L*-rhamnopyranosyl-(1→2) -*O*-[*β*-*D*-glucopyranosyl-(1→6)]-*β*-*D*-glucopyranoside	$C_{45}H_{72}O_{17}$	*L.candidum*	[41]
80	(25*R*)-26-(*β*-*D*-glucopyranosyl) -22-methoxyfurost-5-en-3*β*-yl *O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→6)-*β*-*D*-glucopyranoside	$C_{52}H_{86}O_{23}$	*L.candidum*	[41]
81	(23*S*,25*R*)-23-hydroxyspirost-5-en-3*β*-yl *O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→6)]-*β*-*D*-glucopyranoside	$C_{45}H_{72}O_{18}$	*L.candidum*	[41]
82	(25*R*,26*R*)-26-methoxyspirost-5-ene-3*β*-ol 3-*O*-{*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[6-*O*-acetyl-*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside}	$C_{48}H_{76}O_{19}$	*L.speciosum*	[42]
83	(25*R*,26*R*)-26-methoxyspirost-5-en-3*β*-ol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside	$C_{40}H_{64}O_{13}$	*L.dauricum*	[43]
84	(25*R*,26*R*)-26-methoxyspirost-5-en-3*β*-ol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{46}H_{74}O_{18}$	*L.dauricum*	[43]
85	(25*R*,26*R*)-26-methoxyspirost-5-en-3*β*-ol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*α*-*L*-arabinopyranosyl-(1→3)]-*β*-*D*-glucopyranoside	$C_{45}H_{72}O_{17}$	*L.dauricum*	[43]
86	(25*R*)-spirost-5-en-3*β*-ol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2) -*O*-[*α*-*L*-arabinopyranosyl-(1→3)]-*β*-*D*-glucopyranoside (lililancifoloside A)	$C_{44}H_{70}O_{16}$	*L.dauricum*	[43]

No.	Compound name	Molecular formula	Source	[Ref.]
87	(25*R*)-3*β*,17*α*-dihydroxy-5*α*-spirostan-6-one 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2) -*O*-[*α*-*L*-arabinopyranosyl-(1→3)]-*β*-*D*-glucopyranoside	$C_{44}H_{70}O_{18}$	*L.dauricum*	[43]
88	(20*R*,22*R*)-3*β*,20,22-trihydroxy-5*α*-cholestan-6-one 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2) -*β*-*D*-glucopyranoside	$C_{33}H_{56}O_{10}$	*L.dauricum*	[43]
89	(25*R*,26*R*)-26-ethoxyspirost-5-ene-3*β*-ol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	~	*L.candidum*	[44]
90	(25*R*,26*R*)-3*β*-{*β*-*D*-glucopyranosyl-(1→4)-[*α*-*L*-rhamnopyranosyl-(1→2)]-*β*-*D*-glucopyranosyloxy}-26-ethoxy-spirost-5-ene	$C_{47}H_{76}O_{18}$	*L.candidum*	[44]
91	(25*R*)-spirost-5-ene-3*β*-ol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-xyl-(1→3)]-*β*-*D*-glucopyranoside (ophiopogonin D')	$C_{44}H_{70}O_{16}$	*L.lancifolium*	[45]
92	(25*R*)-spirost-5-en-3*β*-ol 3-*O*-*β*-*D*-glucopyranoside (lilioglycoside A /trillin)	$C_{33}H_{52}O_{8}$	*L.regale*	[46]
93	(25*R*)-spirost-5-en-3*β*-ol 3-*O*-[*O*-*L*-rhamnopyranosyl-(1→2)-*D*-glucopyranoside] (lilioglycoside *D* / prosapogenin A of dioscin)	$C_{39}H_{62}O_{12}$	*L.regale*	[46]
94	(25*R*)-furost-5-en-3*β*,22*α*,26-triol 3,26-di-*O*-*β*-*D*-glucopyranoside (lilioglycoside K)	$C_{39}H_{64}O_{14}$	*L.regale*	[46]
95	(25*R*)-furost-5-en-3*β*,22*α*,26-triol 3-*O*-[O-*L*-rhamnopyranosyl-(1→2)-D-glucopyranoside 26-*O*-*β*-*D*-glucopyranoside (lilioglycoside N)	$C_{45}H_{74}O_{18}$	*L.regale*	[46]
96	(25*R*)-furost-5-en-3*β*,22*α*,26-triol 3-*O*-{[O-*L*-rhamnopyranosyl-(1→2)][O-D-glucopyranosyl-(1→3)]-*β*-*D*-glucopyranoside} 26-*O*-*β*-*D*-glucopyranoside (lilioglycoside R)	$C_{51}H_{84}O_{23}$	*L.regale*	[46]
97	(25*R*)-spirost-5-en-3*β*-ol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→6)]-*β*-*D*-glucopyranoside	~	*L.lancifolium*	[47]
98	(25*R*)-spirost-5-en-3*β*-ol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*α*-*L*-xylopyranosyl-(1→3)]-*β*-*D*-glucopyranoside	~	*L.lancifolium*	[48]
99	(25*R*)-27-*O*-(3-hydroxy-3-methylglutaroyl) -spirost-5-en-3*β*,27-diol 3-*O*-*α*-*L*-rha-(l→2) -*β*-*D*-glucopyranoside (brownioside /lilioglycoside F)	$C_{45}H_{70}O_{17}$	*L.brownii*	[49]
100	(25*R*)-26-*O*-(*β*-*D*-glucopyranosyl)-furost-5-en-3*β*,22*α*,26-triol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2) [*α*-*L*-xylopyranosyl-(1→3)]-*β*-*D*-glucopyranoside	$C_{50}H_{82}O_{22}$	*L.longiflorum*	[50]
101	(25*R*)-26-*O*-(*β*-*D*-glucopyranosyl)-furost-5-en-3*β*,22*α*,26-triol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2) [*α*-*L*-arabinopyranosyl-(1→3)]-*β*-*D*-glucopyranoside	$C_{50}H_{82}O_{22}$	*L.longiflorum*	[50]
102	(25*R*)-26-*O*-(*β*-*D*-glucopyranosyl)-furost-5-en-3*β*,22*α*,26-triol 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*α*-*L*-xylopyranosyl-(1→3)-*β*-*D*-glucopyranoside	~	*L.longiflorum*	[51]

No.	Compound name	Molecular formula	Source	[Ref.]
103	(25*R*)-26-[(*β*-*D*-glucopyranosyl) oxy]-22*α*-hydroxyfurost-5-en-3*β*-yl O-*α*-*L*-arabinopyranosyl-(1→4) -*O*-[*β*-*D*-glucopyranosyl-(1→3)]-*O*-[*α*-L rhamnopyranosyl-(1→2)]-*β*-*D*-glucopyranoside	$C_{56}H_{92}O_{27}$	*L.pumilum DC*	[52]
104	(25*R*)-26-[(*β*-*D*-glucopyranosyl) oxy]-22*α*-hydroxyfurost-5-en-3*β*-yl O-*α*-*L*-arabinopyranosyl-(1→3) -*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*O*-[*α*-L rhamnopyranosyl-(1→2)]-*β*-*D*-glucopyranoside	$C_{56}H_{92}O_{27}$	*L.pumilum DC*	[52]
105	(25*R*)-26-[(*β*-*D*-glucopyranosyl) oxy]-furosta-5,20(22) -dien-3*β*-yl O-*α*-*L*-arabinopyranosyl-(1→4)-*O*-[*β*-*D*-glucopyranosyl-(1→3)]-*O*-[*α*-L rhamnopyranosyl-(1→2)]-*β*-*D*-glucopyranoside	$C_{56}H_{90}O_{26}$	*L.pumilum DC*	[52]
106	(25*R*)-26-[(*β*-*D*-glucopyranosyl) oxy]-furosta-5,20(22) -dien-3*β*-yl O-*α*-*L*-arabinopyranosyl-(1→3)-*O*-[*β*-*D*-glucopyranosyl-(1→4)］-*O*-[*α*-L rhamnopyranosyl-(1→2)]-*β*-*D*-glucopyranoside	$C_{56}H_{90}O_{26}$	*L.pumilum DC*	[52]
107	(20*R*,22*R*)-3*β*,20,22-tetrahydroxy-5*α*-cholestan-6-one 3-*O*-*β*-*D*-glucopyranoside (tenuifolioside A)	$C_{33}H_{56}O_{10}$	*L.tenuifolium*	[53]
108	(20*R*,22*R*)-3*β*,14*α*,20,22-tetrahydroxy-5*α*-cholestan-6-one 3-*O*-*β*-*D*-allopyranoside (tenuifolioside B)	$C_{39}H_{66}O_{14}$	*L.tenuifolium*	[53]
109	26-*O*-*β*-*D*-glucopyranosylnuatigenin 3-*O*-*α*-*L*-rhamnopyranosyl-(1 → 2)-*O*-[(4-*O*-methyl)-*β*-*D*-glucopyranosyl-(1→6)-*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside (lilyasiaside A)	$C_{58}H_{94}O_{28}$	*L.Asiatic hybrids*	[54]
110	26-*O*-*β*-*D*-glucopyranosylnuatigenin 3-*O*-*β*-*D*-glucopyranosyl-(1 → 4)-*α*-*L*-rhamnopyranosyl-(1→2)-(6-acetoxyl)-*β*-*D*-glucopyranoside (lilyasiaside B)	$C_{53}H_{84}O_{24}$	*L.Asiatic hybrids*	[54]
111	*O*-*β*-*D*-glucopyranosylnuatigenin 3-*O*-*β*-*D*-glucopyranosyl-(1 → 2)-*O*-[*β*-*D*-glucopyranosyl-(1→6)]-*β*-*D*-glucopyranoside (lilyasiaside C)	$C_{51}H_{82}O_{24}$	*L.Asiatic hybrids*	[54]

Table 2 Polysaccharides of *Lilii Bulbus*

No.	Compound name	Composition	Molar ratio	Mol. Wt./ kDa	[Ref.]
1	LP2-1	L-Rha, D-Ara, Glu acid, Gal acid	1.88 : 2.13 : 1.00 : 2.50	8520.0	[4]
2	LLP-1	Glu, Man, Gal, Ara, Rha, Glu acid, Gal acid	9.56 : 4.57 : 43.05 : 37.11 : 1.96 : 2.56 : 1.09	2250.0	[9]
3	LLP-2	Glu, Man, Gal, Ara, Rha, Glu acid, Gal acid	20.23 : 2.02 : 30.20 : 32.47 : 3.88 : 2.63 : 1.57	2020.0	[9]
4	LLP-3	Glu, Man,Gal, Ara,Rha,Glu acid, Gal acid	68.55 : 3.04 : 11.60 : 8.75 : 4.77 : 1.76 : 1.53	2080.0	[9]
5	BHP-1	Glu, Man	5.9 : 2.0	193.0	[13]

No.	Compound name	Composition	Molar ratio	Mol. Wt./ kDa	[Ref.]
6	BHP-2	Glu, Man, Gal, Ara	8.3 : 1.0 : 1.5 : 1.1	88.6	[13]
7	LDP	D-Man,D-Glc,D-Gal	10 : 19 : 1	5170.0	[55]
8	LBPS-1	Glu	~	3020.0	[56]
9	LBP-1	Glu	~	30.5	[57]
10	LLPS-1	Glu, Man	1 : 2	350.5	[58]
11	LLPS-2	Glu, Man	1 : 1	403.3	[58]
12	LLPS-3	Glu, Man, Ara, Gla	2 : 1 : 2 : 2	146.2	[59]
13	LP1	Glu, Man	1 : 2.46	79.4	[60]
14	LP2	Glu, Man, Gal, Ara, GalA	1 : 2.6 : 0.73 : 1.8 : 0.84	18.15	[60]
15	BHP	Glu, Man, Ara,	~	75.0	[61]
16	LPR	Glu, Man	2.9 : 3.3	51.2	[62]
17	LLP-1A	Glu, Man	1 : 1.77	78.61	[63]

Table 3 Flavonoids in *Lilii Bulbus*

No.	Compound name	Molecular formula	Mol. Wt.	Ref.
1	Rutin	$C_{27}H_{30}O_{16}$	610	[63]
2	Dihydroquercetin	$C_{15}H_{12}O_7$	304	[63]
3	Kaempferol	$C_{15}H_{10}O_6$	286	[63]
4	Dihydromyricetin	$C_{15}H_{12}O_8$	320	[63]
5	Eriodictyol	$C_{15}H_{12}O_6$	288	[63]
6	(+)-Catechin	$C_{15}H_{14}O_6$	290	[63]
7	(-)-Epi-catechin	$C_{15}H_{14}O_6$	290	[63]
8	Cyanidin 3-rutinoside(keracyanin chloride)	$C_{27}H_{31}O_{15}$	595	[63]
9	Quercetin	$C_{15}H_{10}O_7$	302	[64]
10	Myricetin	$C_{15}H_{10}O_8$	318	[64]
11	Phloridzin	$C_{21}H_{24}O_{10}$	436	[64]
12	Rutinoside	$C_{28}H_{32}O_{16}$	624	[64]
13	Kaempferol-3-*O*-rutinoside	$C_{27}H_{30}O_{15}$	594	[65]
14	Isorhamnetin-3-*O*-rutinoside	$C_{28}H_{32}O_{16}$	624	[65]

Table 4 Alkaloids in *Lilii Bulbus*

No.	Compound name	Molecular formula	Mol. Wt.	[Ref.]
1	(22*R*,25*R*)-spirosol-5α-3β-0-[α-*L*-rhamnopyranosyl-(1→2)]-β-*D*-glucopyranoside	~	~	[12]
2	(22*R*,25*R*)-spirosol-5α-3β-0-α-*L*-rhamnopyranosyl-(1→2)-[β-*D*-glucopyranoside-(1→4)]-β-*D*-glucopyranoside	$C_{45}H_{75}NO_{16}$	885	[12]
3	Berberine	$C_{20}H_{18}NO_4$	336	[32]

No.	Compound name	Molecular formula	Mol. Wt.	[Ref.]
4	Adenodine	$C_{10}H_{13}N_5O_4$	267	[32]
5	*β*1-solamargine	$C_{39}H_{63}NO_{11}$	721	[34]
6	Solasodine-3-0-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranosyl-(1→4)]-*β*-*D*-glucopyranoside	$C_{45}H_{73}NO_{16}$	883	[34]
7	Jatropham	$C_5H_7NO_2$	131	[34]
8	Jatropham 5-0-*β*-*D*-glucopyranoside	$C_{11}H_{17}NO_7$	275	[34]
9	Jatropham 5-0-[0-*β*-*D*-glucopyranoside-(1→3)]-*β*-*D*-glucopyranoside	$C_{17}H_{27}NO_{12}$	437	[34]
10	(22*R*,25*R*)-spirosol-5-en-3*β*-yl 3-0-*α*-*L*-rhamnopyranosyl-(1→2)-*β*-*D*-glucopyranoside	$C_{39}H_{63}NO_{11}$	721	[34]
11	(22*R*,25*R*)-spirosol-5-en-3*β*-yl 3-0-*α*-*L*-rhamnopyranosyl-(1→2)-0-[*β*-*D*-glucopyranoside-(1→4)]-*β*-*D*-glucopyranoside	$C_{45}H_{74}NO_{16}$	885	[34]
12	5,5'-Oxydi(3-methyl-3-pyrrolin-2-one)	$C_{10}H_{12}N_2O_3$	208	[34]
13	1-(3-Methyl-2-oxo-3-pyrrolinyl)-4-methyl-5-oxo-2-pyrrolidineic acid	$C_{11}H_{14}N_2O_4$	238	[34]
14	(+)-5-0-methyljatropham	$C_6H_9NO_2$	127	[34]
15	(22*R*,25*R*)-spirosol-5-en-3*β*-yl 3-0-*α*-*L*-rhamnopyranosyl-(1→2)-0-[6-0-acetyl-*β*-*D*-glucopyranoside-(1→4)]-*β*-*D*-glucopyranoside	$C_{47}H_{76}NO_{17}$	927	[50]
16	Colchicine	$C_{22}H_{25}NO_6$	399	[66]
17	*β*-lumicolchicine	$C_{22}H_{25}NO_6$	399	[67]
18	*β*2-solamargine	$C_{39H63}NO_{11}$	721	[67]
19	2'-deoxyadenosine	$C_{10}H_{13}N_5O_3$	251	[68]
20	Jatropham 5-0-(6-0-p-couaroyl-*β*-*D*-glucopyranoside)	$C_{20}H_{23}NO_9$	421	[69]
21	Solanidine 3-*O*-*α*-*L*-rhamnopyranosyl-(1→2)-*O*-[*β*-*D*-glucopyranoside-(1→4)]-*β*-*D*-glucopyranoside	$C_{45}H_{73}NO_{15}$	867	[70]

Table 5 Phenylpropenoid glycerides of *Lilii Bulbus*

No.	Compound name	Molecular formula	Mol. Wt.	[Ref.]
1	1-0-caffeoylglycerol	$C_{12}H_{14}O_6$	254	[12]
2	1-0-feruloylglycerol	$C_{13}H_{16}O_6$	268	[12]
3	1-0-p-coumaroylglycerol	$C_{12}H_{14}O_5$	238	[12]
4	1,2-0-diferuloylglyccerol	$C_{23}H_{24}O_9$	444	[12]
5	1,3-0-diferuloylglyccerol	$C_{23}H_{24}O_9$	444	[12]
8	1-0-feruloy-3-0-p-coumaroylglycerol	$C_{22}H_{22}O_8$	414	[12]
10	1-0-p-coumaroyl-2-0-feruloylglyccerol	$C_{22}H_{22}O_8$	414	[12]

No.	Compound name	Molecular formula	Mol. Wt.	[Ref.]
37	Threo-1-(4'-hydroxy-2'-methoxyphenyl) -2-(2'',4''-dihydroxyphenyl) -1,3-propanediol-4'-*O*-*β*-*D*-glucopyranoside	~	~	[31]
39	1-*O*-p-coumaroyl-3-*O*-feruloylglycerol	$C_{22}H_{22}O_8$	414	[34]
28	Protocatechualdehyde	$C_7H_6O_3$	138	[47]
29	Salicylic acid	$C_7H_6O_3$	138	[47]
9	1-0-feruloy-2-0-p-coumaroylglycerol	$C_{22}H_{22}O_8$	414	[53]
17	(2*R*)-1-*O*-*β*-*D*-glucopyranosyl-2-*O*-p-coumaroylglycerol (regaloside H)	$C_{18}H_{24}O_{10}$	400	[53]
18	(2*R*)-1-*O*-*β*-*D*-glucopyranosyl-2-*O*-p-coumaroyl-3-*O*-acetylglycerol (regaloside I)	$C_{20}H_{26}O_{11}$	442	[53]
36	6-*O*-acetyl-3,6'-*O*-diferuloylsucrose	~	~	[53]
38	l-*O*-coumaroyl-2-*O*-p-feruloylglycerol	~	~	[53]
30	3,4-Dihydroxybenzaldehyde	$C_7H_6O_3$	138	[55]
31	2-Hydroxybenzoic acid	$C_7H_6O_3$	138	[55]
22	Chlorogenic acid	$C_{16}H_{18}O_9$	354	[63]
27	Gallic acid	$C_7H_6O_5$	170	[64]
15	(2*S*)-1-*O*-caffeoyl-2-*O*-*β*-*D*-glucopyranosylglycerol (regaloside K)	$C_{18}H_{24}O_{11}$	416	[68]
45	4,4'-di-*O*-acetyl-3,6'-di-*O*-feruloylsucrose	~	~	[68]
46	3,4'-di-*O*-p-coumaroylsucrose	~	~	[68]
47	3,6'-di-*O*-p-coumaroylsucrose	~	~	[68]
48	2'-*O*-acetyl-3,4'-di-*O*-p-coumaroylsucrose	~	~	[68]
49	3'-*O*-acetyl-3,4'-di-*O*-p-coumaroylsucrose	~	~	[68]
51	(2*S*)-1-*O*-p-methoxycinnamoyl-2-*O*-*β*-*D*-glcopyranosylferuloyl (methylregaloside D)	$C_{19}H_{26}O_{10}$	414	[68]
52	(2*S*)-1-*O*-cinnamoyl-2-*O*-*β*-*D*-glcopyranosylferuloyl (regaloside J)	$C_{18}H_{24}O_9$	384	[68]
53	(2*R*)-1-*O*-*β*-*D*-glucopyranosyl-2-*O*-caffeoylglycerol (regaloside L)	$C_{18}H_{24}O_{11}$	416	[68]
6	1,3-0-di-p-coumaroylglycerol	$C_{21}H_{20}O_7$	384	[71]
7	1-0-caffeoyl-3-0-p-coumaroylglycerol	$C_{21}H_{20}O_8$	400	[71]
11	(2*S*)-1-*O*-p-coumaroyl-3-*O*-*β*-*D*-glucopyranosylglycerol (regaloside A)	$C_{18}H_{24}O_{10}$	400	[72]
13	(2*S*)-1-*O*-caffeoyl-3-*O*-*β*-*D*-glucopyranosylglycerol (regaloside C)	$C_{18}H_{24}O_{11}$	416	[72]
14	(2*S*)-1-*O*-p-coumaroyl-2-*O*-*β*-*D*-glucopyranosylglycerol (regaloside D)	$C_{18}H_{24}O_{10}$	400	[72]
20	regaloside F	$C_{19}H_{26}O_{12}$	430	[72]
21	regaloside G	$C_{19}H_{26}O_{11}$	430	[72]
26	3,6'-0-diferuloylsucrose	$C_{32}H_{38}O_{17}$	694	[72]

No.	Compound name	Molecular formula	Mol. Wt.	[Ref.]
12	(2*S*)-1-*O*-p-coumaroyl-2-*O*-*β*-*D*-glucopyranosyl-3-*O*-acetylglycerol (regaloside B)	$C_{20}H_{26}O_{11}$	442	[73]
43	Acetylregaloside C	$C_{20}H_{26}O_{12}$	458	[73]
16	(2*S*)-1-*O*-caffeoyl-2-*O*-*β*-*D*-glucopyranosyl-3-*O*-acetylglycerol	~	~	[74]
19	regaloside E	$C_{20}H_{26}O_{12}$	458	[75]
33	4-*O*-acetyl-3,6'-*O*-diferuloylsucrose	$C_{34}H_{40}O_{18}$	736	[75]
44	(2)-hydroxy-3-*O*-p-coumaroyl-1,2-propanedicarboxylic acid	$C_{14}H_{14}O_8$	310	[75]
23	Eugenol 4-0-*α*-*L*-rhamnopyranosyl-(1→6)-*β*-*D*-glucopyranoside	~	~	[76]
24	2,6-dimethoxy-4-(prop-2-enyl) phenyl 0-*α*-*L*-rhamnopyranosyl-(1→6）-*β*-*D*-glucopyranoside	~	~	[76]
25	2,6-dimethoxy-4-(prop-2-enyl) phenyl 0-*β*-*L*-glucopyranosyl-(1→6）-*β*-*D*-glucopyranoside	~	~	[76]
32	3-*O*-feruloylsucrose	$C_{22}H_{30}O_{14}$	518	[77]
34	4'-*O*-acetyl-3,6'-*O*-diferuloylsucrose	~	~	[78]
35	3-*O*-feruloyl-6'-*O*-[4-*O*-*β*-*D*-glcopyranosylferuloyl］sucrose	~	~	[78]
50	(2*S*)-1-*O*-p-methoxycinnamoyl-3-*O*-*β*-*D*-glcopyranosylferuloyl (methylregaloside A)	$C_{19}H_{26}O_{10}$	414	[78]
40	P-coumaric acid	$C_9H_8O_3$	164	[79]
41	Ferulic acid	$C_{10}H_{10}O_4$	194	[79]
42	Caffeic acid	$C_9H_8O_4$	180	[80]

Table 6 Other chemical components of *Lilii Bulbus*

No.	Compound name	Molecular formula	Mol. Wt.	[Ref.]
1	*D*-galactose	$C_6H_{12}O_6$	180	[53]
2	Methyl-*β*-*D*-fructofuranside	$C_7H_{14}O_6$	194	[53]
3	Methyl-*α*-*D*-mannopyranodside	~	~	[53]
4	Adenosine	$C_{10}H_{13}N_5O_4$	267	[69]
5	Docosanoic acid	$C_{22}H_{44}O_2$	340	[69]
6	Methyl-*α*-*D*-glucopyranoside	$C_7H_{14}O_6$	194	[69]
7	Hexacosan-1-ol	$C_{26}H_{54}O$	382	[69]
8	2,4,6-Trichlorol-3-methyl-5-methoxy-phenol-1-*O*-*β*-*D*-Glucopyranosyl-(1→6)-*β*-*D*-glucopyranoside	$C_{20}H_{27}Cl_3O_{12}$	587	[76]
9	4-Chlorol-5-hydroxyl-3-methyl-phenol-1-*O*-*α*-*L*-rhamnopyranosyl-(1→6)-*β*-*D*-glucopyranoside	$C_{19}H_{27}ClO_{11}$	489	[76]

No.	Compound name	Molecular formula	Mol. Wt.	[Ref.]
10	*N*-butyl-*β*-*D*-fructopyranoside	$C_{10}H_{20}O_6$	236	[81]
11	*β*-sitosterol	$C_{29}H_{50}O$	414	[82]
12	P-hydroxybenzaldehyde	$C_7H_6O_3$	122	[82]
12	Eicosannoic acid	$C_{20}H_{40}O_2$	312	[82]
14	Heneicosanoic acid	$C_{21}H_{42}O_2$	326	[82]
15	*N*- tetratriacontanol	$C_{34}H_{70}O$	494	[82]
16	Vanilic acid	$C_8H_8O_4$	168	[83]
17	Syringic acid	$C_9H_{10}O_5$	198	[83]
18	*N*-docosane acid	$C_{22}H_{44}O_2$	340	[84]
19	Nonacosanol	$C_{29}H_{60}O$	424	[84]

Table 7 Biological activity of *Lilii Bulbus*

Biological Activities	Extract methods/Compounds Name	Models and Effective Concentrations	[Ref.]
Anti-inflammatory	Ethanol extract	LPS-treated RAW 264.7 cells (10, 20, 50, and 100 μg/ml)	[85]
	Methanol extract	Xylene-induced mice ear edema modal (2 g/kg)and carrageenan induced mice hind paw edema model (4 g/kg)	[86]
	Water extract	Cigarette smoke esposed mouse modaol (50 mg/kg)	[87]
	Ethanol extract	LPS-treated RAW 264.7 cells (75-600 μg/mL)	[8]
	1-*O*-feruloyl-2-*O*-p-coumaroylglycerol, 1,3-*O*-diferuloylglycerol	LPS-treated RAW 264.7 cells (IC_{50}: 9.12±0.72μM,12.01±1.07 μM)	[87]
Antioxidant	LLP1, LLP2, LLP3	DPPH, hydroxyl and superoxide radicals, Fe2+ (0.125-5.0 mg/mL)	[9]
	LP2-1	DPPH, hydroxyl radicals, and Fe^{2+} (0.5-5.0 mg/mL)	[4]
	LPR	DPPH,hydroxyl, and ABTS free radicals (0.125-4.0 mg/mL)	[62]
	BHP-1	DPPH and hydroxyl radicals (IC_{50} :1.6,1.5 mg/mL)	[13]

Anti-tumor	BHP-2	DPPH and hydroxyl radicals (IC_{50} :1.4,1.3 mg/mL)	[13]
	1,3-0-di-p-coumaroylglycerol、1-0-caffeoyl-3-0-p-coumaroylglycerol	DPPH and ABTS free radicals,Fe2+ (0.01-6.00 mM,0.05-3.00 mM,and 0.3-1.5 mM)	[71]
	LBP-1	Lewis lung carcinoma (LBP-1:50, 100, and 200 mg/kg)	[57]
	Colchicine	A549 cells (Colchicine:2.5 nM)	[88]
	Methanol extracts	MCF-7 cells (IC_{50}:1.06 μg/mL)	[89]
	Atropuroside A	GSC-7901, K562, and SPCA-1 cells (IC_{50} = 4,41,93 μmol L–1)	[30]
	Extracted by hot water, water-soluble polysaccharide from *Lilium davidii* var. unicolor Cotton (LP60-1)	The inhibition rate of HepG2 cell was 81.45% (IC_{50} = 123.8 mμg/mL)	[10]
	Regaloside A	Cell counting kit-8 assay,and western blotting (400 μMCORT,5-100 μM-regaloside A)	[90]
Antidepressant	Saponins、1 time heat reflux extraction with 70 % ethanol for 3 h	Immobility time of mice in the tail suspension and forced swimming tests and antagonism of reserpine-induced hypotherma	[11]
Hypoglycemic	Aqueous extract	Ovariectomized (OVX) mice (1.8 g/kg aqueous extract of lily bulb)	[91]
	Steroidal glycoside rich L. brownii fraction(SGL)、ultrasonic extracted with 500 mL of MeOH for three times, each 2 h	SGL could increase glucose consumption in HepG2 cells and 3T3-L1 adipocytes by 1.61- and 2.13-fold (SGL:50 μg/mL)	[12]
	Water extraction and alcohol precipitation, Alkali washing lily polysaccharide Ⅲ	models of type Ⅰ diabetes induced by streptozotocin (100 and 200 mg /(kg·d))	[92]
	Gradient alcohol precipitation, four polysaccharides (BLP-30, BLP-50, BLP-70, and BLP-80)	α-amylase and α-glucosidase (BLP-70>BLP-80>BLP-30>BLP-50, BLP-50>BLP-70>BLP-80>BLP-30)	[93]
	Lilium lancifolium polysaccharides (LLP)	Streptozotocin (STZ)-induced diabetic mice (LLP:200 mg/kg)	[94]
Antibacterial	Steroidal glycosides (two steroidal glycoalkaloids and three furostanol saponins)	An *in vitro* plate assay (Botrytis cinerea Pers. Fr.)	[95]
	Colchicine	Escherichia coli, Staphylococcus aureus, Bacillus subtilis, Albicans bacteria, and Sarcina lutea (Minimal inhibitory concentration: 0.20, 0.20, 0.24, 0.16, and 0.20 mg/mL)	[96]

BHP-1、BHP-2	*B. cereus*, *P. putida*, *M. luteus*, and *K. pneumonia* with inhibition zones of 13 mm, 16 mm, 17 mm, 23 mm *C.* and 12 mm, 17 mm, 19 mm, 20 mm (16 mg/mL)	[13]
Water extract	Staphylococcus aureus, Escherichia coli,Pseudomonas aeruginosa, Micrococcus luteus, Enterococcus faecalis, Aspergillus flavus (Minimal inhibitory concentration: 50, 25, 12.5, 25, 50,and 25 mg/mL)	[97]
Ethanol extract	Staphylococcus aureus, Escherichia coli,Pseudomonas aeruginosa, Micrococcus luteus,Enterococcus faecalis, and Aspergillus flavus (Minimal inhibitory concentration: 6.25, 12.5, 25, 6.25, 25,and 12.5 mg/mL)	[97]
Ethyl acetate extract	Staphylococcus aureus, Escherichia coli, Pseudomonas aeruginosa,Micrococcus luteus, Enterococcus faecalis, and Aspergillus flavus (Minimal inhibitory concentration: 3.13, 50, 6.25, 50, 50, and 12.5 mg/mL)	[98]

Journals Analysis

A total of 449 English-language journals were published in *Lilii Bulbus* literature related to research from 1998-2023, and the journals with more than thirty publications were, respectively, “FOOD CHEM”, “J AGR FOOD CHEM”, “CARBOHYD POLYM”, “PHYTOCHEMISTRY”, “MOLECULES”, “ETHNOPHARMACOLOGY”, “INT J BIOL MACROMOL”, “ANAL BIOCHEM”. The papers from these journals are related to *Lilii Bulbus*, as shown in Table 8. In this study, the distribution of journals by field of expertise was analyzed using journal biplot overlay as shown in Figure 2. The left side represents citing journals, and the right side represents cited journals. The dual-map overlay revealed three major citation paths (Figure 2). These paths indicate that that journals in Veterinary/Animal/Science frequently cite those in Environmental/Toxicology/Nutrition (z = 3.79, f = 3618);

Journals in Veterinary/Animal/Science also cite those in Molecular/Biology/Genetics (z = 2.16, f = 2268);

Journals in Molecular/Biology/Immunology cite those in Molecular/Biology/Genetics (z = 1.99, f = 2124).

Here, z denotes citation strength, and f denotes citation frequency.

Table 8 The top eight journals on *Lilii Bulbus* research

Journal name	Country/Area	Count	IF 2023
Food Chem	England	71	8.8
Journal of Agricultural and Food Chemistry	United States	57	6.4
Carbohyd Polym	England	55	11.2
Phytochemistry	England	51	3.8
Molecules	Switzerland	49	4.6
Journal of Ethnopharmacology	Ireland	46	5.4
International Journal of Biological Macromolecules	Netherlands	44	8.2
Analytical Biochemistry	United States	31	2.9

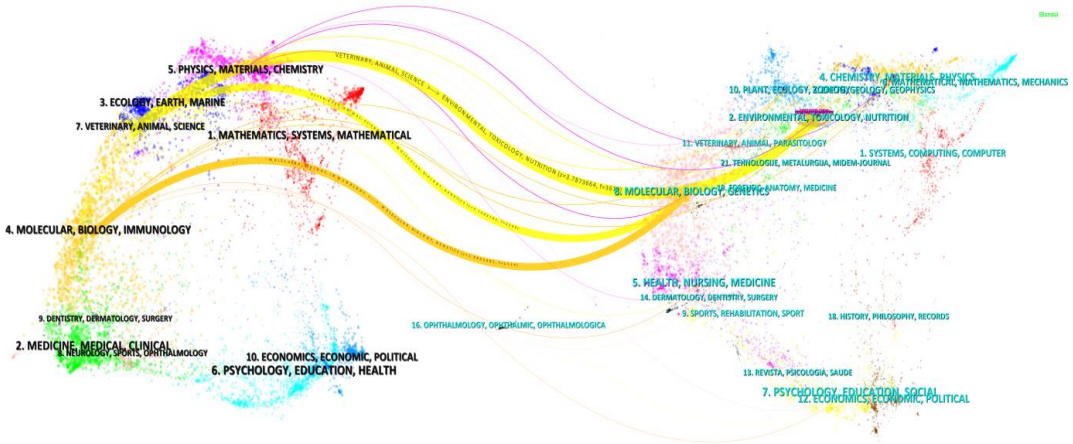


Figure 2 The dual-map overlay of journals on *Lilii Bulbus* research

Document Co-Cited Analysis

The analysis results of co-cited articles are shown in Figure 3. A network map with 279 nodes and 691 links of document co-citations was generated to form a cooperative knowledge map, with the network density being 0.0178 from English literature. The top 30 co-cited references on *Lilii Bulbus* research are shown in Table 9. Among them, the highest number of citations is Li, WM published in J AGR FOOD CHEM for the characterization of Lanzhou lily (*Lilium davidii* var.unicolor) polysaccharides and determination of their associated antioxidant activity (16 co-citations). In this study, the authors used a Box-Behnken design to optimize the ultrasound-assisted extraction of Lanzhou lily polysaccharides and the Lanzhou lily polysaccharide was identified as alpha-(1 → 6)-d-glucan. Consequently, the effects of both potential antioxidant and proliferative activity of trypsin are significant [24].

Table 9 The top 30 co-cited references on *Lilii Bulbus* research

Source	Author	Frequency	[Ref.]
Food Sci Hum Well	Wang PY	4	[1]
Carbohyd Polym	Zhou J	7	[100]
Appl Biochem Biotech	Li H	6	[101]
J Agr Food Chem	Wang FX	6	[102]
Postharvest Biol Tec	Munafo JP	5	[103]
J Food Biochem	Huang DJ	5	[104]
J Agr Food Chem	Munafo JP	5	[105]
Int J Biol Macromol	Li H	4	[106]
Planta	Abbas F	3	[107]
Int J Biol Macromol	Hui HP	3	[108]
Spectrochim Acta A	Li YE	3	[109]
Int J Biol Macromol	Huang JH	3	[110]

J Food Biochem	Kan J	3	[111]
Int J Biol Macromol	Haladova M	3	[112]
Food Chem	Hui HP	9	[13]
J Agr Food Chem	Li WM	16	[24]
Phytochemistry	Mimaki Y	3	[36]
Molecules	Gao J	8	[4]
Phytochemistry	Satou T	3	[40]
Int Immunopharmacol	Munafo JP	10	[50]
Int J Food Prop	Chen ZG	6	[58]
Int J Biol Macromol	Pan GF	10	[59]
J Food Sci	Hui HP	7	[62]
Acs Omega	Murray AF	3	[75]
Nat Prod Rep	Sim WS	5	[8]
J Ethnopharmacol	Kwon OK	4	[85]
J Ethnopharmacol	Wang TT	7	[86]
Carbohyd Polym	Ma T	8	[87]
Int J Biol Macromol	Huang H	7	[99]

Keywords Co-Occurrence Analysis

Keywords in the literature are important information for authors to express their research themes, research methods and core ideas. The frequency of keywords is positively correlated with research hotspots. By counting the frequency of keywords and conducting keyword co-occurrence analysis, it can further explore the trends in the development of research themes [25].

By constructing a visual mapping of keyword co-occurrence analysis from the literature in English, it can be seen from the figure that 295 nodes and 844 connecting lines between nodes are constructed by the keywords. The top 8 keywords include antioxidant activity, saponin, physicochemical property, glycoside, constituent, *Lilium lancifolium*, extract, and purification. They can be classified into 5 categories: chemical components, biological activities, species origin, quality evaluation, methods of extraction and purification, as shown in Table 10. The most frequent categories are chemical components and biological activities, with a frequency of 35, indicating that researchers are most concerned with the chemical components of *Lilii Bulbus*.

The keywords and keyword frequencies of *Lilii Bulbus* research literature in Chinese can be obtained by using Citespace software, and the 286 keywords from the literature can be classified into five categories: quality evaluation, chemical components, species origin, biological activities, and extraction methods, as shown in Table 10. The most frequent category is quality evaluation, with a frequency of 43, indicating that the research focus is on quality evaluation, while the chemical components of *Lilii Bulbus* share the same research focus as that of the literature in English.

Table 10 Keyword of literature on *Lilii Bulbus* research

Classification	Keywords (frequency)	Frequency
Keyword of English literature on *Lilii Bulbus* research		
Chemical components	Constituents (10), glycosides (9), saponins (8), and steroidal saponins (8)	35
Biological activities	Antioxidant activity (29) and antioxidant (6)	35
Species origin	*Lilium lancifolium* (10), *Lilium longiflorum thumb* (7), and *Lanzhou lily* (6)	23
Quality evaluation	*Lilii Bulbus* (22) and chemical characterization (6)	18
Methods of extraction and purification	Purification (8) and extract (4)	12
Keyword of Chinese literature on *Lilii Bulbus* research		
Quality evaluation	*Lilii Bulbus* (28), content determination (8), and regaloside b (7)	43
Chemical components	Chemical constituents (8), lily polysaccharide (19), steroidal saponins (6), and total saponins (6)	39
Species origin	*Lilium lancifolium thunb* (15), *Lilium brownii* (6), and *Lanzhou lily* (4)	25
Biological activities	Antioxidant activity (10)	10
Extraction methods	Extraction process (4) and extraction technology (4)	8

Keyword Clustering Analysis

In this study, 295 keywords from the literature in English are clustered automatically using the log likelihood ratio (LLR) test algorithm, which is divided into 11 clustering modules. The clustering modules are numbered from #0 to #11, as follows #0 steroidal glycoside, #1 phenolic compounds, #2 isolation, #3 *Lilium bulbs*, #4 antifungal activity, #5 *Lilium brownii* var. viridulum, #6 volatile organic compounds, #7 Lanzhou lily, #8 anti inflammatory, #9 *Lilium candidum*, #10 diabetes mellitus, #11 starch, as shown in Figure 4A. The clusters interact with each other and can be divided into 5 categories: #0 steroidal glycoside, #1 phenolic compounds, #6 polysaccharides and #11 starch are chemical components; #5 *Lilium brownii* var. viridulum, #7 *Lanzhou lily* and #9 *Lilium candidum* are species origin; #4 antifungal activity and #8 anti inflammatory are pharmacological and medicinal effects; #2 isolation refers to extraction and separation; and #11 starch is related to quality evaluation.

Based on the results of clustering, a timeline diagram was drawn and shown in Figure 4B. The time span of each cluster and the connection between different clusters can be visualized through the timeline diagram, which illustrates the development of lily research [26]. The horizontal coordinates indicate the year of publication, the vertical coordinates indicate the cluster number and name, and each node represents a keyword; and the higher the frequency, the larger the node. The results show that cluster #0, steroidal glycoside, represents the longest research span in the field.

In this study, 286 keywords from literature in Chinese are clustered automatically using the log likelihood ratio (LLR) test algorithm, which is divided into 12 clustering modules. The clustering modules are numbered from #0 to #11, as follows #0 *Lilium lancifolium* Thunb., #1 *lily polysaccharide*, #2 intrinsic free sugars, #3 *lily bulbus*, #4 total flavonoids, #5 anti-oxidant activity, #6 total saponins, #8 bulbus lilii, #9 isolation and purification, and #11 regalosible B, as shown in Figure 5A. The clusters interact with each other and can be divided into five categories: #1 lily polysaccharide, #4 total flavonoids, #7 total saponins, #11 regaloside B, and #2 intrinsic free sugars are chemical components; #0 *Lilium lancifolium* Thunb. represents species origin; #3 lily bulbus, and #8 bulbus lilii relate to quality evaluation; #5 anti-oxidant activity pertains to pharmacological and medicinal effects; #9 isolation and purification involves extraction and separation. Clusters #6 and #10 were not generated in the analysis,

likely due to low keyword frequency or merging with adjacent clusters.

Based on the results of clustering, a timeline diagram was drawn and shown in Figure 5B. The results show that cluster #0 *Lilium lancifolium* Thunb. represents the longest research span in the field, while cluster #2 intrinsic free sugars; #3 lily bulbus; and #4 total flavonoids represent the research spans from 2009, 2010, and 2011 to the present, respectively.

Keyword clustering analysis is conducive to obtaining the distribution of *Lilii Bulbus*-related research and hotspots. The values of modularity (Q) and silhouette (S) are indices to measure the effect of clustering. In general, the value of Q is within the interval of [0, 1), and Q > 0.3 indicates that the clustering results are significant. The larger value of Q, the more suitable for clustering, while S > 0.5 is reasonable. The larger value of S and closer to 1, the stronger homogeneity of keywords in each cluster [27]. Q is 0.7879 in English literature research; the clustering is very significant. S is between 0.9292. Q is 0.8782 in Chinese literature research; the clustering is more significant, and S is 0.9713, making the clustering confidence is high.

Burst Topics

Burst words are keywords that appear with sudden frequency over a brief time. Burst strength is the intensity of a sudden increase in frequency in a short period of time. Burst words and burst strength can reflect the trend of the research frontier [28]. The start time and end time of the mutation are indicated by "begin" and "end" respectively, and "strength" refers to the burst strength. The higher the burst strength, the stronger the emergence, resulting in a greater the impact. The number of red squares shows the annual distribution of the burst of the burst word.

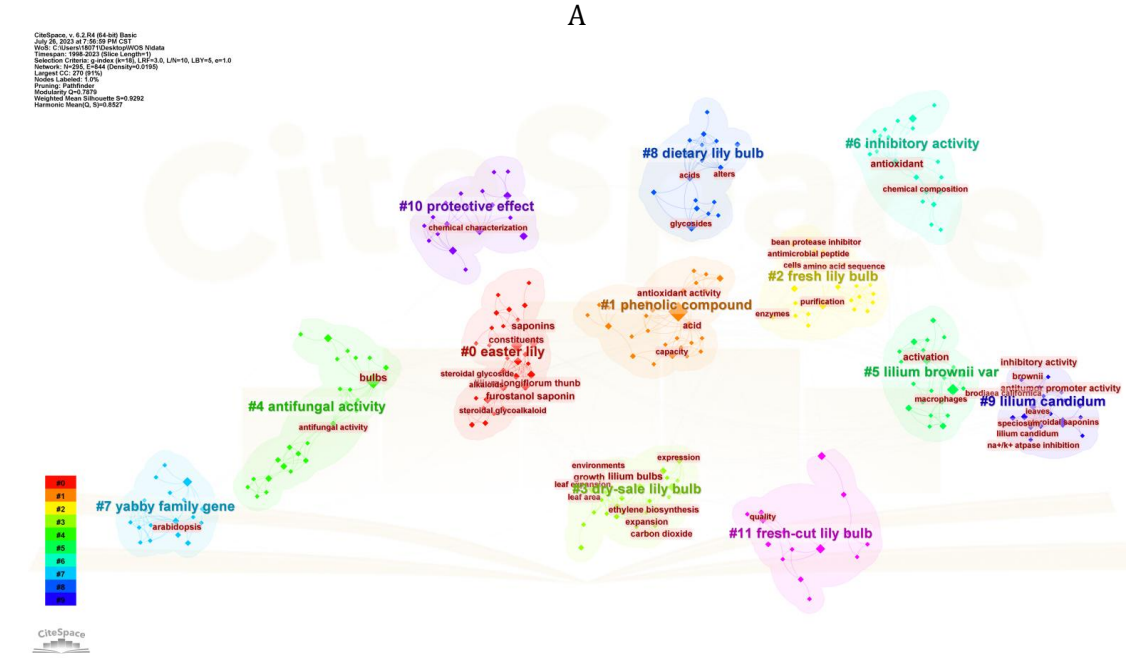
The burst words mapping of English literature on *Lilii Bulbus* research is shown in Figure 6A. The top 18 burst words are *Lilium candidum, Lilium longiflorum* Thunb., furostanol saponin, steroidal glycoside, steroidal glycoalkaloid, saponins, Easter lily, glycosides, steroidal saponins, extracts, antioxidant activity, activation, *Lilium lancifolium*, gene expression, lily bulbs, Lanzhou lily, plants, and purification. Overall, the burst strength of the top 18 burst words ranged from 1.69 to 2.69, while endurance time span ranged from 1 to 9 years. The burst words mapping clearly presents the evolution of lily research hotspots in the temporal dimension. The results show that the English literature before 2016; mainly focused on saponins, while the research hotspots after 2016 focused on biological activities and methods of extraction and purification.

The results of the burst words mapping are shown in the Figure 6B. The top 10 burst words in Chinese literature are *Lilium brownii*, bulbus *Lilii, Lilii* polysaccharides, lily polysaccharide, steroidal saponins, antioxidant activity, content determination, *Lilii Bulbus*, regaloside B, and cluster analysis. Overall, the burst strength of the top 10 burst words ranged from 1.8 to 4.94, while the endurance time span ranged from 1 to 8 years. The burst words mapping clearly presents the evolution of *Lilii Bulbus* research hotspots in the temporal dimension. The results show that the Chinese literature before 2013 focused on component analysis, while the research hotspots after 2015 concentrated on pharmacological mechanisms; and quality evaluation is a current hotspot.

Country/Region Publication Output (1998–2023)

To profile the global research landscape of *Lilii Bulbus*, the 143 English-language records retrieved from the Web of Science Core Collection were analyzed with CiteSpace 6.2.R4 (Country node, g-index k = 25, Pathfinder pruning). As shown in Table 11, scholars affiliated with China (PRC) contributed 98 publications, corresponding to 68.5 % of the corpus and far exceeding any other country. The United States ranked second with 12 papers (8.4 %), followed by South Korea (8; 5.6 %), Japan (6; 4.2 %), and Italy (5; 3.5 %). This distribution confirms China's dominant role in generating English-language research on *Lilii Bulbus* over the past 25 years.

A



B

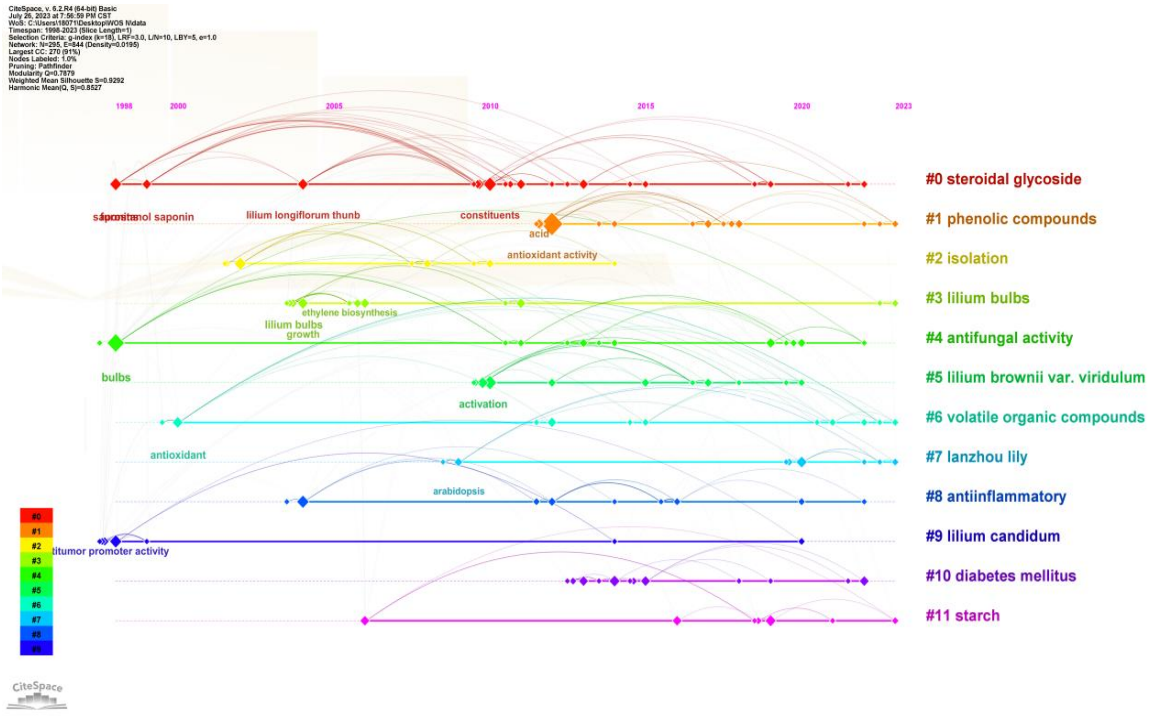
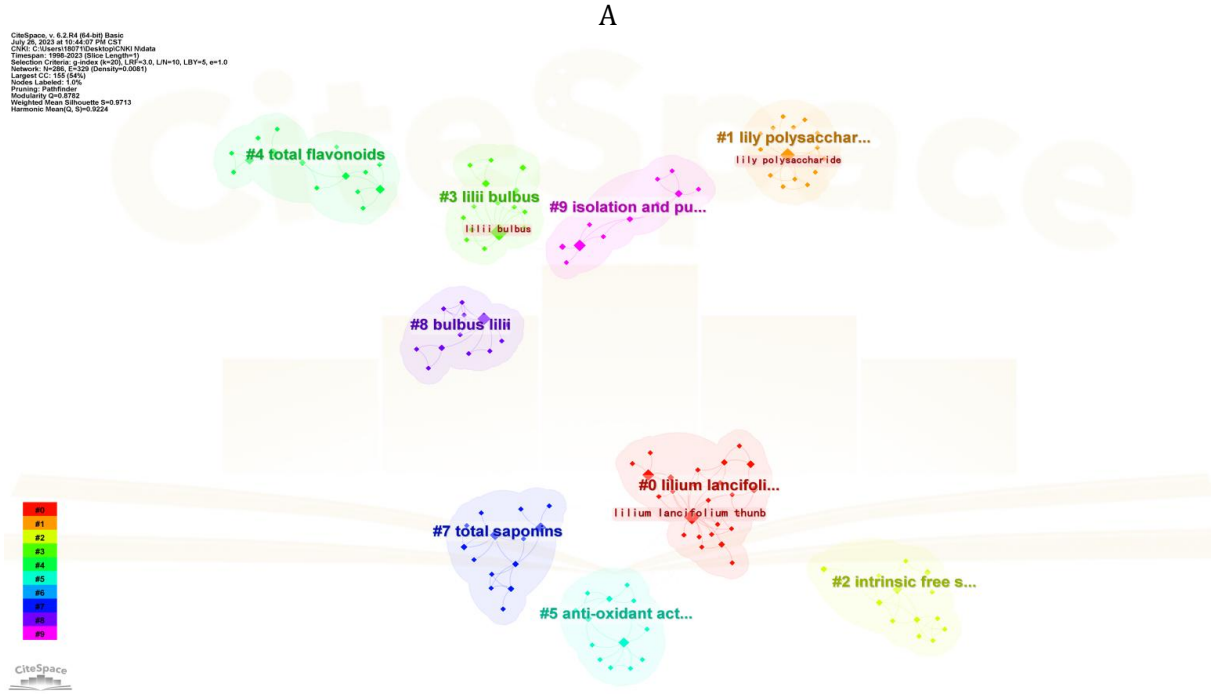


Figure 4 The cluster map of keywords (A) and a timeline diagram (B) of English literature on *Lilii Bulbus* research

A



B

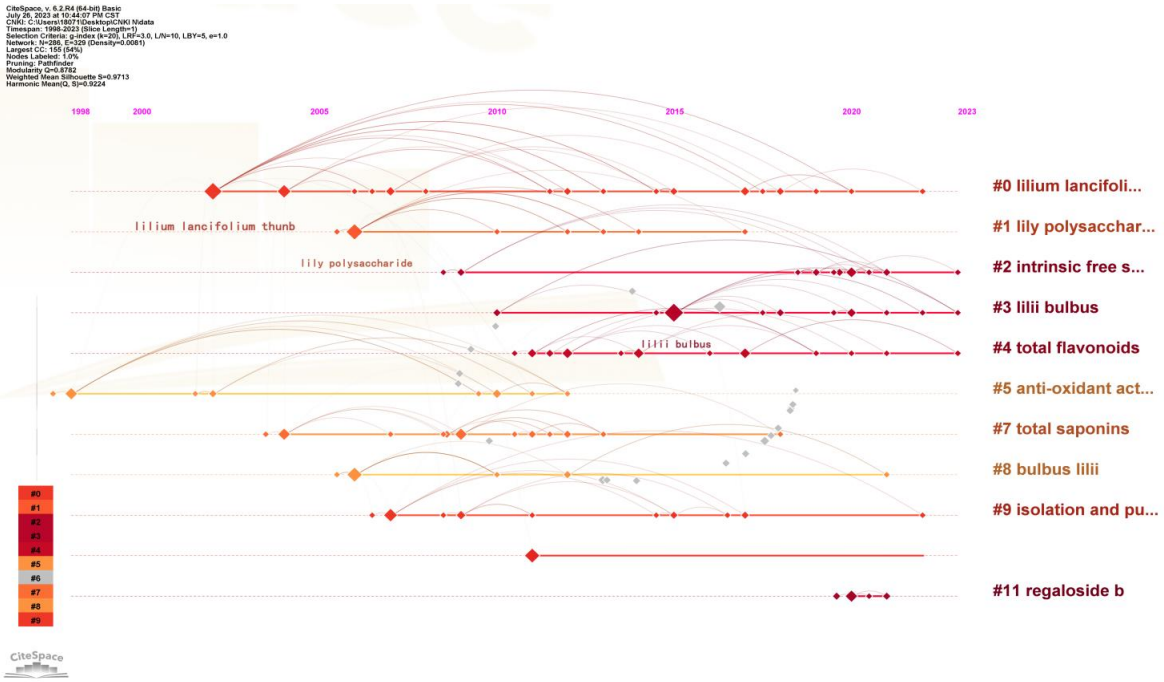


Figure 5 The cluster map of keywords (A) and a timeline diagram (B) of Chinese literature on *Lilii Bulbus* research

A

Keywords	Year	Strength	Begin	End	1998 - 2023
lilium candidum	1998	1.9	1998	2001	
lilium longiflorum thunb	2004	2.11	2004	2015	
furostanol saponin	1999	1.9	2010	2013	
steroidal glycoside	2010	1.79	2010	2011	
steroidal glycoalkaloid	2010	1.79	2010	2011	
saponins	1998	1.77	2010	2015	
easter lily	2011	1.78	2011	2015	
glycosides	2004	2.4	2012	2015	
steroidal saponins	1998	1.82	2014	2015	
extracts	2016	2.03	2016	2019	
antioxidant activity	2012	2.69	2017	2023	
activation	2010	1.69	2017	2018	
lilium lancifolium	2010	2.97	2019	2023	
gene expression	2013	1.84	2019	2020	
lily bulbs	2019	1.69	2019	2023	
lanzhou lily	2020	2.22	2020	2023	
plants	2006	2.05	2020	2021	
purification	2002	2.37	2021	2023	

B

Keywords	Year	Strength	Begin	End	1998 - 2023
lilium brownii	2004	1.95	2004	2012	
bulbus lilii	2006	3.67	2006	2012	
lily polysaccharides	2006	1.8	2006	2013	
lily polysaccharide	2006	3.58	2009	2014	
steroidal saponins	1998	2.6	2010	2012	
antioxidant activity	2011	2.1	2015	2017	
content determination	2004	1.71	2019	2021	
lilii bulbus	2015	4.94	2020	2023	
regaloside b	2020	3.68	2020	2021	
cluster analysis	2020	1.67	2020	2023	

Figure 6 Top 18 keywords with the strongest citation bursts in English literature (A) and top 10 keywords with the strongest citation bursts in Chinese literature (B)

Table 11 Top five countries/regions by publication count

Rank	Country/Region	Publications	Percentage (%)
1	China (PRC)	98	68.5
2	United States	12	8.4
3	South Korea	8	5.6
4	Japan	6	4.2
5	Italy	5	3.5

Conclusion

235 chemical components (111 steroidal saponins, 17 polysaccharides, 14 flavonoids, 21 alkaloids, 53 phenylpropenoid glycerides, and 19 other chemical components) and the biological activities from extraction methods, compounds, models, effective concentrations were systematically summarized. CiteSpace was employed for scientometric analysis in the fields of chemical components and biological activities research of *Lilii Bulbus*. The results show that the

majority of literature was published in PRC. The core journal was Journal of FOOD CHEM (impact factor = 8.8). The keywords can be classified into five categories: chemical components, biological activities, species origin, quality evaluation, methods of extraction and purification. The results indicate that the English literature before 2016; mainly focused on saponins, while the research hotspots after 2016 concentrated on biological activities and methods of extraction and purification. Similarly, the Chinese literature before 2013 emphasized component analysis, while the research hotspots after 2015 focused on pharmacological mechanisms with quality evaluation being a current hotspot. The identification and assessment of chemical components (like steroidal glycosides) of *Lilii Bulbus* and their biological activities (such as anti-inflammatory and antioxidant) are the current research foci for *Lilii Bulbus* research. To sum up, *Lilii Bulbus*, which is homologous in medicine and food, has great potential in the treatment of cancer and inflammation and it can also be used as a cosmetic raw material of natural origin to deeply explore the efficacy of cosmetics and increase the added value of products. The present scientometric study provides the hot topics and dynamic frontiers of the development of *Lilii Bulbus* research over the decades using quantitative and qualitative methods, and this review can provide a reference for subsequent research directions focusing on *Lilii Bulbus*.

Authors' Contributions

Software, M.H.; resources, Y.C.; data curation, Y.C.; writing—original draft preparation, Y.H.; writing—review and editing, Y.C.; visualization, M.H.; data analysis, X.W; supervision, and S.Z.; All authors have read and agreed to the published version of the manuscript.

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